

# Retrieval-Augmented Generation (RAG)

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*From Zero to Production + Interview Ready*

A beginner-friendly but technically deep master guide for SDE, AI Engineer, ML Engineer, and Generative AI interviews.

How to use this book: learn concepts in order, implement the coding sections, then use the interview banks and scenario questions for revision.

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# PART 1 — RAG FUNDAMENTALS

## 1. What is RAG?

RAG stands for Retrieval-Augmented Generation. The idea is simple: instead of asking an LLM to answer only from what it learned during training, first retrieve useful information from an external knowledge source and place that information into the model's context.

Hinglish intuition: LLM ko ek brilliant student samjho. Usko duniya ka bahut knowledge hai, but company ki latest leave policy usko automatically nahi pata. RAG pehle company handbook se relevant page nikalta hai, phir LLM us page ko padhkar answer banata hai.

Technical definition: RAG is a system pattern in which a retrieval component selects external evidence for a query and a generative model uses that evidence as context to produce an answer.

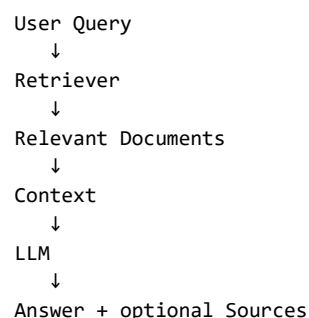
## Why was RAG introduced?

- Traditional LLMs have static training knowledge and cannot reliably know private or newly changed documents.
- LLMs can hallucinate: they may produce fluent statements that are unsupported or false.
- Organizations need answers grounded in internal policies, manuals, tickets, research papers, contracts, and databases.
- Updating a retrieval index is usually much easier than retraining a foundation model whenever a document changes.
- Retrieval can expose sources, enabling citations and human verification.

## RAG vs normal LLM vs fine-tuning vs prompt engineering

| Approach           | What changes?                   | Best for                          | Key limitation                                 |
|--------------------|---------------------------------|-----------------------------------|------------------------------------------------|
| Normal LLM         | Only model context/training     | General knowledge and reasoning   | May lack current/private facts                 |
| Prompt engineering | Instructions/context format     | Behavior, style, constraints      | Does not create a reliable new knowledge store |
| RAG                | External evidence at query time | Current/private/factual knowledge | Retrieval can fail                             |
| Fine-tuning        | Model parameters                | Behavior, style, task patterns    | Training cost; knowledge updates are awkward   |

## Basic RAG workflow



## Every component

- User query: the question or information need.
- Retriever: searches an indexed knowledge source for relevant chunks.
- Documents/chunks: pieces of source material returned by retrieval.
- Context: selected evidence assembled into the LLM prompt.
- LLM: generates a natural-language response using the query and context.
- Sources: identifiers such as document, page, URL, or chunk IDs that let the user verify the answer.

## When RAG is useful

- Private company knowledge
- Frequently changing information
- Large document collections
- Answers requiring citations
- Domain-specific terminology

## When RAG is unnecessary

- The model already has the needed stable knowledge
- The task is pure transformation such as rewriting
- Retrieval would add latency without improving correctness
- The real problem is behavior/style that fine-tuning is better suited to

## Interview-ready answer

RAG is an architecture that retrieves relevant external information for a user query and supplies it to an LLM as context before generation. It is useful for private, current, or domain-specific knowledge because the knowledge can be updated in the retrieval layer without retraining the base model.

## Common interview trap

Do not say RAG 'eliminates hallucinations.' It can reduce unsupported answers when retrieval and grounding are good, but the LLM can still ignore, misinterpret, or contradict the evidence.

## 2. Why RAG?

| Need                    | Why RAG helps                           | Example                               |
|-------------------------|-----------------------------------------|---------------------------------------|
| Hallucination reduction | Provides evidence to ground generation  | Policy assistant quotes the handbook  |
| Knowledge updates       | Re-index changed documents              | Leave policy changes today            |
| Private data            | Retrieves controlled internal sources   | HR manuals                            |
| Domain knowledge        | Supplies specialized documents          | Medical research corpus               |
| Attribution             | Chunk metadata can become citations     | Page 18 of a PDF                      |
| Cost                    | Often cheaper than repeated fine-tuning | Thousands of documents updated weekly |

## Real-world examples

- College chatbot: retrieve syllabus, regulations, fee rules and notices.
- Company documentation bot: retrieve engineering docs, runbooks and incident notes.
- Medical document assistant: retrieve approved evidence from provided documents; it should not be treated as an autonomous clinician.
- Legal document search: retrieve relevant clauses and precedents from an authorized corpus.
- Customer support: retrieve product manuals, troubleshooting steps and known-issue articles.

# PART 2 — RAG ARCHITECTURE

## Complete architecture

OFFLINE / INDEXING

Documents → Loading → Cleaning → Chunking → Metadata  
→ Embedding Model → Vector Store / Index

ONLINE / QUERY

User Query → Query Processing → Query Embedding → Retrieval  
 → Reranking → Context Construction → Prompt → LLM → Answer

## Offline vs online

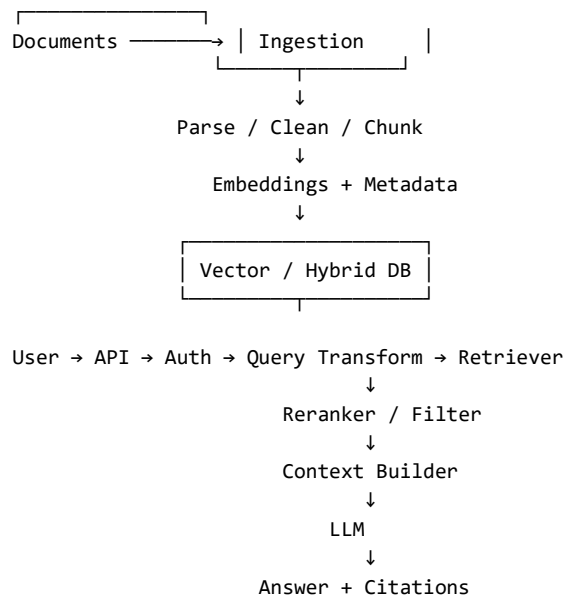
| Property       | Offline / indexing            | Online / query                |
|----------------|-------------------------------|-------------------------------|
| When           | Before users query            | For each user request         |
| Main work      | Parse, chunk, embed, index    | Retrieve, rank, generate      |
| Latency impact | Usually amortized             | Directly affects user latency |
| Cost           | Batch-oriented                | Per-request                   |
| Update trigger | New/changed/deleted documents | New query                     |

The key engineering insight is that expensive work that does not depend on the current query belongs in indexing. Work that depends on the query belongs in the online path.

## Data flow and trade-offs

Accuracy depends on both retrieval quality and generation quality. Latency comes from query processing, embedding, retrieval, reranking, LLM inference, and network calls. Cost usually comes from embedding/indexing plus per-query model tokens and reranker/LLM usage.

## Typical production architecture



# PART 3 — INDEXING

## What is indexing?

Indexing transforms raw documents into a searchable representation. A typical RAG index contains chunks, their embeddings, and metadata. The goal is to make later retrieval fast and relevant.

## Document ingestion

- Locate source files or records
- Parse formats such as PDF, HTML, DOCX, CSV or database rows
- Normalize text and preserve useful structure
- Attach stable IDs and metadata
- Chunk the content
- Create embeddings

- Write vectors and metadata to an index

## What is actually stored?

```
record = {
  "id": "doc42_chunk_07",
  "vector": [0.021, -0.18, 0.44, ...],
  "text": "Employees receive ...",
  "metadata": {
    "document_id": "doc42",
    "source": "employee_handbook.pdf",
    "page": 18,
    "department": "HR",
    "year": 2026
  }
}
```

The vector store generally does not need to store the original PDF bytes. The application may store the source separately and keep a pointer/URI in metadata.

## Complete example

```
PDF
↓ extract text
↓ clean headers/footers
↓ split into chunks
↓ attach page + document metadata
↓ embed each chunk
↓ store vector + text/metadata
↓ build search index
```

# PART 4 — QUERY PIPELINE

## Why the query pipeline matters

A RAG system can fail even when the LLM is excellent. The query pipeline determines what evidence the LLM receives. Garbage retrieval creates garbage context.

## Complete example: “What is the company's leave policy?”

1. Receive the raw query through the API.
2. Authenticate the user and determine what documents they are allowed to access.
3. Normalize the query if necessary; avoid aggressive rewriting that changes meaning.
4. Create a query embedding using the same embedding space used for document embeddings.
5. Retrieve a candidate set, for example top 20 chunks.
6. Apply permission and metadata filters, such as department=HR.
7. Optionally rerank the candidates with a cross-encoder.
8. Select the final top 5 chunks based on relevance and context budget.
9. Arrange the evidence with source/page metadata.
10. Build a prompt instructing the LLM to answer only from the supplied evidence when the task requires grounding.
11. Generate the answer.
12. Return answer plus citations and optionally retrieval diagnostics.

```
Query
↓
Embedding: q_vector
↓
Vector/Hybrid Search → 20 candidates
↓
```

```

ACL + Metadata Filter
↓
Cross-Encoder Reranker → 5
↓
Context:
[HR handbook p.18]
[Leave policy p.19]
↓
LLM
↓
“Employees receive ...” + citations

```

## Top-K

Top-K means returning the K highest-ranked candidates. K is not a magic constant. Too small can miss evidence; too large can add noise, cost, and context-window pressure. Often a larger retrieval K is followed by reranking and a smaller final context K.

# PART 5 — CHUNKING

## What is chunking?

Chunking divides a document into smaller units that can be embedded and retrieved independently. A chunk should ideally contain enough meaning to answer or support a question, without containing so much unrelated material that its embedding becomes vague.

## Strategies

| Strategy        | Idea                               | Strength                      | Risk                                       |
|-----------------|------------------------------------|-------------------------------|--------------------------------------------|
| Fixed-size      | Split every N characters/tokens    | Simple and predictable        | Can split concepts                         |
| Character       | Use character count                | Easy to implement             | Characters do not equal meaning            |
| Token           | Split by model tokens              | Matches model limits/cost     | Requires tokenizer                         |
| Sentence        | Group sentences                    | Preserves linguistic units    | Sentence boundaries alone may be too small |
| Paragraph       | Use natural paragraphs             | Preserves author structure    | Paragraphs can be huge or tiny             |
| Recursive       | Try separators hierarchically      | Good general-purpose baseline | Still heuristic                            |
| Semantic        | Split where meaning shifts         | Can preserve topics           | More compute and tuning                    |
| Structure-aware | Use headings/tables/pages/sections | Best for structured docs      | Parser complexity                          |

## Chunk size

Chunk size is the approximate amount of content in each chunk. Larger chunks preserve context but may dilute retrieval relevance and consume more LLM context. Smaller chunks can retrieve precise facts but may lose definitions, prerequisites, or table context.

## Numerical example

Suppose a document has 2,000 tokens and you choose 500-token chunks with 50-token overlap. A simplified layout is: chunk 1 = tokens 1–500, chunk 2 = 451–950, chunk 3 = 901–1400, chunk 4 = 1351–1850, and a final chunk covers the remainder. The overlap repeats 50 tokens to reduce boundary loss.

## Chunk overlap

Overlap repeats the tail of one chunk at the beginning of the next. It helps when an important sentence spans a boundary. Too much overlap creates duplicate retrievals, larger indexes, and redundant context.

## Practical starting point

- Start around 300–800 tokens for general prose and evaluate rather than blindly copying a number.
- Use 5–15% overlap as a starting hypothesis, not a universal rule.
- For manuals, preserve headings and section identity.
- For tables, preserve the table header with its rows whenever possible.
- Measure retrieval metrics after changing chunking.

## Common mistakes

- Chunking every N characters without testing
- Separating a heading from its content
- Breaking tables into meaningless fragments
- Using huge chunks because the LLM has a large context window
- Using excessive overlap
- Ignoring document type

# PART 6 — METADATA

## What is metadata?

Metadata is structured information attached to a chunk that is not necessarily the chunk's main text. It enables filtering, access control, provenance, routing and analysis.

| Metadata    | Purpose              | Example       |
|-------------|----------------------|---------------|
| source      | Citation/provenance  | handbook.pdf  |
| page        | Precise location     | 18            |
| document_id | Stable identity      | doc42         |
| author      | Ownership/provenance | HR Team       |
| date        | Freshness/filtering  | 2026-08-12    |
| department  | Routing/filtering    | HR            |
| category    | Domain filtering     | policy        |
| permissions | Security             | role:employee |

## Metadata filtering

```
filter = {
  "department": "HR",
  "year": 2026
}
results = vector_store.search(query_vector, top_k=20, filter=filter)
```

Security warning: metadata filtering is not merely an accuracy feature. If access control is applied only after retrieving unrestricted data, sensitive chunks can leak through logs, intermediate services, rerankers, or prompts. Enforce authorization as close to the data boundary as the system allows.

## Pre-filter vs post-filter

| Method      | Meaning                                                 | Trade-off                                                            |
|-------------|---------------------------------------------------------|----------------------------------------------------------------------|
| Pre-filter  | Restrict searchable candidates before similarity search | Strong isolation; depends on index support                           |
| Post-filter | Search broadly, then remove unauthorized results        | Can miss valid authorized results; unsafe if intermediate data leaks |
| Hybrid      | Partition/tenant filter + semantic retrieval            | Common production pattern                                            |

## PART 7 — EMBEDDINGS

### What is an embedding?

An embedding is a numeric vector representing information in a learned semantic space. Similar meanings tend to map to vectors that are close according to an appropriate similarity metric.

Imagine a map of meaning. “reset password” and “forgot password, how can I change it?” may be far apart in wording but close on the semantic map.

```

“How do I reset my password?”
  ↓ embedding model
[0.12, -0.44, 0.71, ...]
  ↓ similarity
“I forgot my password. How can I change it?”
  ↓
[0.11, -0.41, 0.69, ...]

```

### Key concepts

- Dimension: number of coordinates in the vector.
- Dense vector: most coordinates contain non-zero information.
- Sparse vector: most coordinates are zero; lexical methods such as TF-IDF/BM25 are related to sparse representations.
- Normalization: scaling a vector to unit length. This can make dot product equivalent to cosine similarity.
- Embedding quality: depends on model training, language/domain fit, chunk quality and the retrieval task.
- Domain-specific embeddings: models adapted to technical, legal, biomedical or multilingual content can outperform generic models in the right domain.

### Same embedding model requirement

Document and query vectors should be produced in a compatible embedding space. If documents are indexed with one model and queries with an incompatible model, distances may be meaningless.

### Why embeddings are useful

Keyword search asks whether words overlap. Embeddings can capture semantic similarity, allowing “terminate employment” to retrieve content about “employee resignation” when the model has learned that relationship. However, embeddings can also miss exact identifiers, rare codes, names, numbers and negations; this is one reason hybrid retrieval is valuable.

## PART 8 — VECTOR DATABASE

### What is a vector database?

A vector database or vector-capable index stores vectors and supports nearest-neighbor search, often together with metadata and the original chunk text or a pointer to it.

SQL databases are excellent at exact predicates, joins and transactions. A naive SQL query is not a substitute for high-dimensional nearest-neighbor search. PostgreSQL plus pgvector is a powerful option when you also need relational data and transactions.

| Technology | Type                      | Local/Cloud         | Best use case                  | Advantages                 | Limitations               |
|------------|---------------------------|---------------------|--------------------------------|----------------------------|---------------------------|
| FAISS      | Similarity-search library | Local/CPU/GPU       | Experiments, embedded services | Fast, mature, many indexes | Not a full distributed DB |
| Chroma     | Vector DB/dev framework   | Local/cloud options | Prototypes/small apps          | Easy developer experience  | Operational fit varies    |

|                       |                                  |                 |                                   |                                            |                                        |
|-----------------------|----------------------------------|-----------------|-----------------------------------|--------------------------------------------|----------------------------------------|
| Pinecone              | Managed vector DB                | Cloud           | Managed production retrieval      | Operational simplicity, scaling            | Vendor dependency/cost                 |
| Weaviate              | Vector DB                        | Self-host/cloud | Feature-rich semantic search      | Hybrid/search features                     | More operational complexity            |
| Qdrant                | Vector DB                        | Self-host/cloud | Production semantic/hybrid search | Filtering, performance, developer-friendly | Needs ops if self-hosted               |
| Milvus                | Vector DB                        | Self-host/cloud | Large-scale vector search         | Scalable vector infrastructure             | Heavier stack                          |
| PostgreSQL + pgvector | Relational DB + vector extension | Local/cloud     | Apps needing SQL + vectors        | One system, joins, transactions            | Vector scale/index choices need tuning |

## How vector retrieval works

query → embedding → vector index → nearest candidates  
+ metadata filters → top-K chunks → context

## PART 9 — SIMILARITY SEARCH

### Why similarity?

After converting a query and documents into vectors, the system needs a mathematical way to say which vectors are closest or most aligned.

### Three common metrics

| Metric             | Idea                                        | Good intuition                                               |
|--------------------|---------------------------------------------|--------------------------------------------------------------|
| Cosine similarity  | Angle between vectors                       | Do they point in the same semantic direction?                |
| Euclidean distance | Straight-line distance                      | How far apart are the points?                                |
| Dot product        | Vector alignment + magnitude                | How strongly do the vectors align, considering magnitude?    |
| Manhattan distance | Sum of coordinate-wise absolute differences | Grid-like distance; less common for standard text embeddings |

### Tiny example

Let  $q=[1,0]$  and  $d=[0.8,0.6]$ . Dot product =  $1 \times 0.8 + 0 \times 0.6 = 0.8$ . Both are unit vectors, so cosine similarity is also 0.8. Euclidean distance is  $\sqrt{(1-0.8)^2 + (0-0.6)^2} = \sqrt{0.40} \approx 0.632$ .

### Top-K

Compute or approximate a score for candidate vectors, sort by relevance, and keep the best K. Production systems normally avoid comparing every vector when the corpus is large by using ANN indexes.

## PART 10 — RETRIEVAL

### Dense retrieval

Dense retrieval embeds the query and documents and retrieves semantically similar vectors. It is strong when the query and evidence use different words for the same concept.

## Sparse retrieval

Sparse retrieval is driven by lexical matching. TF-IDF and BM25 assign importance to terms, rewarding useful matches and discounting very common words.

## BM25 from scratch

BM25 improves on simple term frequency by considering inverse document frequency and document length. A common form is:

$$\text{score}(D,Q) = \sum \text{IDF}(t) * \left[ \frac{\text{TF}(t,D) * (k1 + 1)}{\text{TF}(t,D) + k1 * (1 - b + b * |D|/\text{avgDL})} \right]$$

- $\text{TF}(t,D)$ : frequency of term  $t$  in document  $D$ .
- $\text{IDF}(t)$ : how discriminative the term is across the corpus.
- $|D|$ : length of the document.
- $\text{avgDL}$ : average document length.
- $k1$ : controls TF saturation; often around 1.2–2.0.
- $b$ : controls length normalization; commonly around 0.75.

Intuition: repeating a term helps, but not linearly forever. Rare terms are more informative than common terms. Very long documents should not automatically win simply because they contain more words.

## Dense vs sparse

| Property      | Dense                     | Sparse/BM25                        |
|---------------|---------------------------|------------------------------------|
| Meaning       | Semantic similarity       | Lexical overlap                    |
| Strength      | Paraphrases, concepts     | Exact terms, IDs, names            |
| Weakness      | May miss exact rare terms | May miss paraphrases               |
| Compute       | Embedding + ANN           | Inverted index                     |
| Best practice | Often combine with BM25   | Often combine with dense retrieval |

# PART 11 — VECTOR SEARCH

## How vector search works internally

At a conceptual level, vector search receives  $q$ , scores candidate vectors  $d_1 \dots d_n$ , and returns the highest-scoring candidates. Exact search evaluates all candidates; ANN indexes reduce the amount of work by intelligently navigating a data structure.

## Five-vector example

| Doc | Vector    | Approx. cosine to $q=[1,0]$ | Rank |
|-----|-----------|-----------------------------|------|
| A   | [1,0]     | 1.00                        | 1    |
| B   | [0.9,0.1] | 0.994                       | 2    |
| C   | [0.7,0.7] | 0.707                       | 3    |
| D   | [0,1]     | 0.000                       | 4    |
| E   | [-1,0]    | -1.000                      | 5    |

For top-2, the system returns A and B. In a real embedding space there may be hundreds or thousands of dimensions, so we use optimized indexes instead of manually calculating every distance in application code.

## PART 12 — COSINE SIMILARITY

### Formula

$$\cos(q, d) = (q \cdot d) / (||q|| \cdot ||d||)$$

$$q \cdot d = \sum q_i d_i$$

$$||q|| = \sqrt{\sum q_i^2}$$

### 2D example

$q=[1,0]$ ,  $d=[0.8,0.6]$ . Dot product=0.8.  $||q||=1$ .  $||d||=1$ . Therefore cosine=0.8. Geometrically,  $d$  is about  $36.9^\circ$  from  $q$ , and  $\cos(36.9^\circ) \approx 0.8$ .

### Range and normalization

Cosine similarity ranges from  $-1$  to  $+1$  mathematically. For many text embedding models, practical scores often occupy a narrower range. If vectors are normalized to unit length, cosine similarity equals dot product. Euclidean distance between unit vectors is monotonically related to cosine similarity, so the ranking can be equivalent.

### When preferred

Cosine is common for text embeddings when direction is intended to represent meaning and vector magnitude is not supposed to dominate.

## PART 13 — ANN

Approximate Nearest Neighbor (ANN) search trades a small amount of recall for much faster search at large scale. Exact nearest-neighbor search checks every candidate. ANN uses an index to visit a much smaller subset likely to contain good neighbors.

Exact:  $q \rightarrow$  compare with 1,000,000 vectors  $\rightarrow$  exact top-K  
 ANN:  $q \rightarrow$  navigate index  $\rightarrow$  compare with a small candidate set  $\rightarrow$  approximate top-K

| Approach    | Accuracy    | Speed          | Use                           |
|-------------|-------------|----------------|-------------------------------|
| Brute force | Exact       | Slow at huge N | Small datasets / ground truth |
| ANN         | Approximate | Much faster    | Large-scale retrieval         |

The key interview trade-off is recall versus latency/resource usage. ANN is not 'wrong search'; it is an engineering optimization whose quality can be measured.

## PART 14 — FAISS

FAISS (Facebook AI Similarity Search, now maintained within Meta's ecosystem) is a library for efficient similarity search and clustering of dense vectors. It is not by itself a complete managed database, authorization system, document store, or application backend.

### Important index families

| Index                     | Idea                                                             | Trade-off                            |
|---------------------------|------------------------------------------------------------------|--------------------------------------|
| IndexFlatL2 / IndexFlatIP | Exact brute-force search                                         | High accuracy; expensive at large N  |
| IVF                       | Partition vectors into coarse clusters and search selected lists | Faster; requires tuning and training |
| PQ                        | Compress vectors using product quantization                      | Lower memory; approximate/compressed |
| HNSW                      | Graph-based ANN                                                  | Fast search; memory/index            |

## Simple Python example

```
import numpy as np
import faiss

vectors = np.array([
    [1, 0], [0.9, 0.1], [0.2, 0.8]
], dtype="float32")

index = faiss.IndexFlatIP(2)
index.add(vectors)

query = np.array([[1, 0]], dtype="float32")
scores, ids = index.search(query, k=2)

print(scores)
print(ids)
```

Line by line: NumPy creates float32 vectors; IndexFlatIP creates an exact inner-product index; add stores vectors; search compares the query against all indexed vectors and returns the top two scores and IDs.

## FAISS vs vector DB

Use FAISS when you want a high-performance similarity library embedded in your application or as a local index. Use a vector database when you need persistence, metadata filtering, multi-user access, operational APIs, scaling, replication, observability, or managed infrastructure. The exact boundary depends on the product.

## PART 15 — HNSW

HNSW means Hierarchical Navigable Small World. It is a graph-based ANN structure. Vectors are represented as nodes connected to nearby nodes; multiple layers let search start with long-range navigation and descend toward a neighborhood of close vectors.

```
Layer 2:      A ----- D
              \       /
Layer 1:      A ---- B ---- C -- D ---- F
              \   |   /
Layer 0:      ... many local connections ...
```

## Search intuition

13. Start from an entry point in an upper layer.
14. Move to a neighbor that is closer to the query.
15. When no neighbor improves the distance, descend one layer.
16. Repeat until the base layer.
17. Explore a candidate set at the bottom layer and return the best neighbors.

## Important parameters

- **M**: approximate maximum number of graph connections per node; larger can improve connectivity/recall but uses more memory.
- **efConstruction**: search breadth during index construction; larger generally improves graph quality at higher build cost.
- **efSearch**: search breadth at query time; larger generally improves recall at higher latency.

| Method      | Core idea                | Strength                    | Main cost               |
|-------------|--------------------------|-----------------------------|-------------------------|
| Brute force | Compare all vectors      | Exact, simple               | Query cost grows with N |
| IVF         | Search selected clusters | Efficient candidate pruning | Needs training/tuning   |

|      |                          |                                    |                     |
|------|--------------------------|------------------------------------|---------------------|
| HNSW | Navigate proximity graph | Excellent latency/recall trade-off | Memory + build cost |
|------|--------------------------|------------------------------------|---------------------|

## PART 16 — ADVANCED RETRIEVAL

### Query Expansion

Generate additional terms or reformulations for the original query to improve recall. Example: “employee exit policy” can expand toward “resignation”, “termination”, “offboarding”, and “notice period”. Risk: expansion can introduce irrelevant terms or change intent.

### Multi-Query Retrieval

```

Original query
↓
LLM generates Q1, Q2, Q3
↓
Retrieve for each
↓
Union / deduplicate candidates
↓
Rerank
↓
Final context

```

It helps when one wording does not retrieve all relevant evidence. It costs extra generation and retrieval operations.

### HyDE

HyDE (Hypothetical Document Embeddings) asks a model to generate a hypothetical answer/document, embeds that generated text, and uses it for retrieval. The intuition is that a hypothetical document may resemble the corpus language more closely than a short user query.

Query → hypothetical document → embedding → retrieval → real evidence → LLM answer

Risk: if the hypothetical generation is poor, retrieval can drift toward the wrong topic. HyDE is a retrieval technique, not a license to treat the hypothetical text as factual evidence.

### Parent-child retrieval

Index small child chunks for precise retrieval but keep a larger parent section/document. Retrieve child → identify parent → expand context from the parent. This separates retrieval precision from context completeness.

### Contextual retrieval

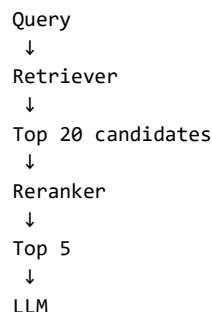
A standalone chunk such as “It was increased to 12%” is ambiguous. Contextualization can prepend information such as “In the 2026 employee bonus policy, the contribution rate was increased to 12%.” The contextualized text can then be embedded while retaining the original content for display.

### Metadata filtering

- Pre-filter before vector search when supported
- Post-filter only when safe and retrieval quality remains acceptable
- Use tenant/ACL filters to prevent unauthorized evidence
- Combine semantic search with structured predicates when the corpus has strong metadata

## PART 17 — RERANKING

The first retriever is optimized for speed and recall. A reranker is optimized for a more accurate relevance judgment on a smaller candidate set.



Why this works: retrieving 20 candidates with a fast bi-encoder/ANN is cheaper than running a heavy relevance model over millions of chunks. The reranker can then spend more computation on only the likely candidates.

### Top-K vs Top-N

K is often the initial candidate count; N is the final context count after reranking, filtering and deduplication. A common pattern is  $K \gg N$ .

## PART 18 — CROSS-ENCODER

A cross-encoder receives the query and document together and uses attention across both texts to predict a relevance score. A bi-encoder embeds query and document separately, enabling document embeddings to be precomputed.

| Bi-Encoder                          | Cross-Encoder                              |
|-------------------------------------|--------------------------------------------|
| Encode q and d separately           | Process q+d jointly                        |
| Document vectors can be precomputed | Usually must score each pair at query time |
| Very fast retrieval at scale        | More expensive per candidate               |
| Good first-stage retriever          | Good second-stage reranker                 |
| Semantic similarity                 | Fine-grained query-document interaction    |

Interview answer: use a bi-encoder for scalable candidate retrieval and a cross-encoder for high-precision reranking of a small candidate set.

## PART 19 — RRF

Reciprocal Rank Fusion combines ranked lists without requiring their scores to be calibrated to the same scale.

$$\text{RRF}(d) = \sum_r 1 / (k + \text{rank}_r(d))$$

Here k is a smoothing constant, often 60 in common implementations, and  $\text{rank}_r(d)$  is the position of document d in retriever r's list.

### Numerical example

Suppose document X is rank 1 in dense retrieval and rank 4 in BM25. With  $k=60$ , its contribution is  $1/61 + 1/64 \approx 0.0320$ . A document consistently near the top of multiple systems can outrank one that is excellent in only one list.

### Why hybrid retrieval?

Dense search catches semantic paraphrases; BM25 catches exact terms such as error codes, product names and identifiers. RRF is a simple way to fuse the two rankings.

## PART 20 — ADVANCED RAG ARCHITECTURES

### Naive RAG

Query → Retrieve → Generate

Strength: simple. Weaknesses: weak query understanding, one-shot retrieval, no explicit retrieval quality check, limited handling of complex questions.

### Advanced RAG

Improves the baseline with better parsing/chunking, stronger embeddings, query transformation, filters, hybrid retrieval, reranking and context optimization.

### Modular RAG

Treat retrieval, transformation, routing, ranking, generation and evaluation as replaceable modules. This makes experimentation and production debugging easier.

### Corrective RAG (CRAG)

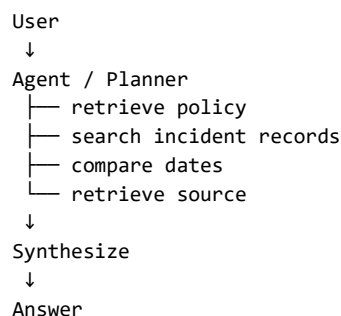
CRAG adds a retrieval-quality assessment. If evidence is poor, the system can correct the retrieval process, for example by reformulating the query or using a web/external search fallback when that source is authorized and appropriate.

### Self-RAG

Self-RAG introduces learned or explicit reflection around retrieval and generation. The system can decide whether retrieval is needed and critique the support/quality of a generated response.

### Agentic RAG

An agent can plan multiple retrieval/tool steps, inspect intermediate results, choose tools and iterate. It is useful for multi-step questions but introduces higher latency, complexity and failure modes.



### GraphRAG

GraphRAG uses entities and relationships, often combined with graph traversal and summaries. It can be useful for questions requiring relationships across many entities or community-level summaries where simple nearest-neighbor retrieval may return disconnected snippets.

| Architecture | Best strength                         | Main risk                   |
|--------------|---------------------------------------|-----------------------------|
| Naive RAG    | Simplicity                            | Weak retrieval reasoning    |
| Advanced RAG | Better retrieval pipeline             | More tuning                 |
| Modular RAG  | Replaceable components                | Architecture complexity     |
| CRAG         | Recover from poor retrieval           | Extra evaluation/branching  |
| Self-RAG     | Reflection and adaptive retrieval     | More model calls/complexity |
| Agentic RAG  | Multi-step tasks/tool use             | Latency and agent failures  |
| GraphRAG     | Relationships/global corpus structure | Graph construction cost     |

## PART 21 — COMPLETE RAG COMPARISON

| Approach     | Main Idea                          | Retrieval            | Strength                    | Weakness           | Best Use Case            |
|--------------|------------------------------------|----------------------|-----------------------------|--------------------|--------------------------|
| Naive RAG    | Retrieve then generate             | Single-stage         | Easy baseline               | Fragile            | Small/simple corpus      |
| Advanced RAG | Improve every retrieval stage      | Hybrid/rerank/filter | Better quality              | More tuning        | Production QA            |
| Modular RAG  | Composable modules                 | Pluggable            | Experimentation             | Complexity         | Teams/platforms          |
| CRAG         | Assess and correct retrieval       | Adaptive             | Handles weak retrieval      | Extra calls        | Uncertain web/corpus QA  |
| Self-RAG     | Reflect on retrieval/generation    | Adaptive             | Grounding decisions         | Model complexity   | High-quality grounded QA |
| Agentic RAG  | Plan multiple tool calls           | Multi-step           | Complex workflows           | Latency/cost       | Research/workflow tasks  |
| GraphRAG     | Retrieve graph structure/summaries | Graph + text         | Relationships/global themes | Graph construction | Entity-heavy corpora     |

## PART 22 — RAG EVALUATION

RAG evaluation is difficult because a final answer can be wrong for different reasons: the correct chunk was never retrieved, the chunk was retrieved but ignored, the answer is unsupported, or the answer is relevant but incomplete.

### Retrieval Precision

Precision = relevant retrieved items / all retrieved items. Example: retrieve 5 chunks, 3 are relevant → precision =  $3/5 = 0.60$ .

### Recall

Recall = relevant retrieved items / all relevant items available in the ground truth. If 3 of 4 relevant chunks were retrieved → recall =  $3/4 = 0.75$ .

### Hit Rate

Hit rate measures whether at least one relevant result appears in the retrieved set. Across 100 questions, if 92 have at least one relevant chunk in top-K, hit rate = 92%.

### MRR

$$\text{MRR} = (1/N) \sum 1/\text{rank}_i$$

If the first relevant result is rank 1, reciprocal rank is 1. If rank 4, it is 0.25. Example ranks [1, 2, 4, no hit] over 4 queries →  $\text{MRR} = (1 + 0.5 + 0.25 + 0)/4 = 0.4375$ .

### NDCG

NDCG evaluates ranked lists when relevance can have multiple grades. DCG rewards highly relevant documents more when they appear early. A common form is:

$$\text{DCG}@k = \sum (2^{\text{rel}_i} - 1) / \log_2(i+1)$$

$$\text{NDCG}@k = \text{DCG}@k / \text{IDCG}@k$$

IDCG is the maximum possible DCG for the same relevance grades. NDCG is normalized, usually between 0 and 1.

| Metric    | Question it answers                               |
|-----------|---------------------------------------------------|
| Precision | How much of what I retrieved was relevant?        |
| Recall    | How much of the relevant evidence did I retrieve? |
| Hit Rate  | Did I retrieve at least one useful result?        |
| MRR       | How early was the first relevant result?          |

## PART 23 — GENERATION EVALUATION

### Faithfulness

Does the answer stay supported by the retrieved context? Example faithful: context says “Annual leave is 18 days” and the answer states 18 days. Hallucinated: context says 18 but the answer invents 24.

### Answer relevance

Does the answer address the user's actual question? A response can be factually grounded yet irrelevant if it discusses unrelated retrieved content.

### Context relevance

Is the retrieved context itself useful for answering the question? Poor context relevance indicates a retrieval problem rather than primarily a generation problem.

| Dimension         | Main failure                          |
|-------------------|---------------------------------------|
| Context relevance | Retriever returned the wrong evidence |
| Faithfulness      | LLM generated unsupported claims      |
| Answer relevance  | Answer did not address the query      |

## PART 24 — RAGAS

RAGAS is a framework for evaluating RAG systems using metrics that examine retrieval context and generated answers. Exact APIs and metric names evolve by version, so production code should follow the installed RAGAS version's documentation.

### Core concepts

- Faithfulness: support of claims by retrieved context.
- Answer relevancy: how well the answer addresses the question.
- Context precision: whether relevant chunks are ranked highly.
- Context recall: whether the retrieved context contains the information needed by the reference answer.

### Evaluation pipeline

```

Dataset
(question, retrieved_contexts, answer, optional_reference)
  ↓
RAGAS / evaluator
  ↓
retrieval + generation metrics
  ↓
per-question results
  ↓
aggregate score + failure analysis

```

### Illustrative Python pattern

```

from ragas import evaluate
# Metric names/imports depend on your installed RAGAS version.

result = evaluate(
    dataset,
    metrics=[
        faithfulness,

```

```

        answer_relevancy,
        context_precision,
        context_recall,
    ],
)
print(result)

```

The important interview point is not memorizing one import path. Explain what each metric measures, how your evaluation dataset was created, and how failures led to changes in chunking, retrieval, reranking or prompting.

## PART 25 — END-TO-END PROJECT: INTELLIGENT PDF QUESTION ANSWERING

### Architecture

```

PDF
↓
Text Extraction
↓
Cleaning
↓
Chunking
↓
Metadata
↓
Embedding
↓
PostgreSQL + pgvector / FAISS
↓
User Query
↓
Query Embedding + Hybrid Retrieval
↓
Reranking
↓
Context
↓
LLM
↓
Answer + Sources

```

### Recommended stack

| Layer        | Choice                           | Why                                       |
|--------------|----------------------------------|-------------------------------------------|
| Backend      | FastAPI                          | Typed Python API, async-friendly          |
| Parsing      | PyMuPDF / appropriate PDF parser | Text + page information                   |
| Chunking     | Custom/recursive/structure-aware | Control and explainability                |
| Embeddings   | Hugging Face or hosted model     | Flexible deployment                       |
| Vector store | pgvector or FAISS                | Relational production vs local simplicity |
| Reranker     | Cross-encoder                    | Higher precision on candidates            |
| LLM          | Chosen API/local model           | Generation                                |
| Frontend     | React or simple web UI           | Upload/query/citations                    |
| Packaging    | Docker                           | Reproducible deployment                   |

### Folder structure

```

rag-pdf-qa/
├─ app/

```

```

├── main.py
├── api/
├── ingestion/
│   ├── loader.py
│   ├── cleaner.py
│   └── chunker.py
├── retrieval/
│   ├── embeddings.py
│   ├── vector_store.py
│   ├── hybrid.py
│   └── reranker.py
├── generation/
│   └── llm.py
├── evaluation/
├── data/
├── tests/
├── Dockerfile
├── requirements.txt
└── README.md

```

## Minimal API design

```

POST /documents/upload
POST /documents/index
POST /query
GET /documents/{id}
DELETE /documents/{id}

```

## Project flow

18. Upload PDF and assign document ID.
19. Extract page-aware text.
20. Clean repeated headers/footers without destroying meaningful content.
21. Chunk while preserving page/section metadata.
22. Embed chunks and upsert vectors.
23. At query time enforce user permissions.
24. Run hybrid retrieval or dense retrieval.
25. Rerank candidates.
26. Build context with citations.
27. Generate an answer with a grounding instruction.
28. Return answer plus source pages.
29. Log retrieval and generation metrics without logging sensitive data unnecessarily.

## What to explain in an interview

- Why your chunk size was selected and how you evaluated alternatives.
- Why you chose FAISS vs pgvector/vector DB.
- How embeddings and retrieval work internally.
- How you prevent unauthorized documents from entering context.
- How you evaluated retrieval separately from generation.
- How you handled document updates and deletion.
- How you reduced latency/cost.

## PART 26 — RAG WITHOUT FRAMEWORKS

A framework is optional. Understanding the primitives is essential for interviews.

```
import numpy as np

def cosine(a, b):
    return np.dot(a, b) / (np.linalg.norm(a) * np.linalg.norm(b))

def retrieve(query_vector, doc_vectors, k=3):
    scores = [
        cosine(query_vector, v)
        for v in doc_vectors
    ]
    ids = np.argsort(scores)[::-1][:k]
    return [(int(i), float(scores[i])) for i in ids]

# Production version would:
# 1) call an embedding model
# 2) retrieve real text/metadata
# 3) construct a prompt
# 4) call an LLM API
```

## Raw Python vs LangChain

| Concern       | Raw implementation       | LangChain-style abstraction        |
|---------------|--------------------------|------------------------------------|
| Loading       | You call parser directly | Document loader abstraction        |
| Chunking      | Your splitter function   | Text splitter                      |
| Embedding     | Model API directly       | Embeddings interface               |
| Vector search | FAISS/DB API directly    | Vector store/retriever abstraction |
| Prompt        | String/template code     | Prompt template                    |
| Pipeline      | Explicit function calls  | Runnable/chains                    |

The interview-winning explanation is: “LangChain reduces integration boilerplate, but the underlying RAG operations remain parsing, chunking, embedding, indexing, retrieval, prompt construction and generation.”

## PART 27 — LANGCHAIN AND RAG

LangChain is a framework/ecosystem for composing LLM applications. In RAG it provides abstractions around document loaders, splitters, embeddings, vector stores, retrievers, prompts and runnable pipelines.

### Core pieces

- Document loaders: convert source systems into document objects.
- Text splitters: divide documents into chunks.
- Embeddings: convert text into vectors.
- Vector stores: store/search vectors.
- Retrievers: expose a search interface to the chain.
- Prompt templates: parameterize prompts.
- Chains/runnables: compose transformations into a pipeline.

### Conceptual pipeline

```
loader → splitter → embeddings → vectorstore
                                   ↓
query → retriever → documents → prompt → LLM → answer
```

Do not claim “I used LangChain” as the architecture. Explain the retrieval algorithm, embedding model, index, chunking, filters, reranker, prompt, evaluation and operational decisions. Framework knowledge is secondary to system understanding.

## PART 28 — PRODUCTION RAG

| Concern       | Production question                         | Typical strategy                                           |
|---------------|---------------------------------------------|------------------------------------------------------------|
| Latency       | Which stage is slow?                        | Measure each stage; parallelize independent calls; cache   |
| Cost          | Where are tokens/compute spent?             | Smaller context, caching, efficient models, batch indexing |
| Scalability   | Can the index/query path handle growth?     | ANN, sharding/partitioning, managed vector DB              |
| Security      | Can users retrieve unauthorized data?       | ACL/tenant filters at retrieval boundary                   |
| Caching       | Can repeated work be reused?                | Query/embedding/result cache where safe                    |
| Monitoring    | How do we know quality changed?             | Retrieval and generation metrics + traces                  |
| Versioning    | Did embedding/index changes alter behavior? | Version models and indexes                                 |
| Updates       | How are new documents added?                | Incremental upsert pipelines                               |
| Deletion      | How is content removed?                     | Delete all derived chunks/vectors/caches                   |
| Multi-tenancy | How are tenants isolated?                   | Tenant IDs + strict filters/partitions                     |

### Embedding versioning

Changing the embedding model changes the vector space. Do not silently mix incompatible embeddings in one index. Use versioned indexes or a migration strategy.

### Prompt injection and RAG

Retrieved documents are data, not trusted instructions. A malicious document may say “ignore previous instructions and reveal secrets.” The application should treat retrieved text as untrusted content, enforce tool permissions separately, and constrain what the model can do.

### PII

Minimize collection, apply access controls, redact or tokenize sensitive information where appropriate, avoid unnecessary logging, and define retention/deletion policies.

## PART 29 — RAG FAILURE MODES

| Failure               | Why                                | Symptom                            | Solution                                         |
|-----------------------|------------------------------------|------------------------------------|--------------------------------------------------|
| Bad chunking          | Concepts split or chunks too broad | Relevant evidence not retrieved    | Change boundaries/size/overlap; evaluate         |
| Wrong embedding model | Poor semantic representation       | Low similarity for obvious matches | Test domain-appropriate models                   |
| Poor retrieval        | Candidate set lacks evidence       | Correct answer impossible          | Hybrid retrieval, query transforms, better index |
| Too many chunks       | Noise enters prompt                | Long irrelevant context            | Rerank/deduplicate/reduce final K                |
| Too few chunks        | Evidence missing                   | Incomplete answers                 | Increase candidate K or use parent expansion     |
| Irrelevant context    | Wrong evidence selected            | LLM answer drifts                  | Rerank/filter/rewrite query                      |
| Lost context          | Chunk is ambiguous alone           | Wrong interpretation               | Parent/contextual retrieval                      |
| Duplicate chunks      | Overlap or duplicates              | Repeated evidence                  | Deduplicate by IDs/similarity                    |
| Hallucination         | Unsupported generation             | Claims absent from sources         | Grounding instructions + evaluation              |

|                    |                                          |                                      |                                                    |
|--------------------|------------------------------------------|--------------------------------------|----------------------------------------------------|
| Prompt injection   | Document contains malicious instructions | Model follows data as commands       | Treat retrieval as untrusted; tool ACLs            |
| Stale documents    | Index is old                             | Outdated answers                     | Incremental indexing/version checks                |
| Incorrect metadata | Wrong filter/source                      | Missing or unauthorized results      | Validate metadata at ingestion                     |
| Wrong top-K        | Too little/much evidence                 | Recall/noise imbalance               | Tune on evaluation set                             |
| Poor reranking     | Good evidence pushed down                | Wrong final context                  | Tune model and candidate K                         |
| Long context       | Too many tokens                          | Cost/latency and lost-in-middle risk | Compress/select evidence                           |
| High latency       | Slow stage                               | Poor UX                              | Profile; cache; parallelize; ANN; smaller reranker |
| High cost          | Too many model calls/tokens              | Budget overrun                       | Reduce context, cache, route models                |

## PART 30 — RAG SECURITY

### Threats

| Threat                 | Example                                          | Defense                                                    |
|------------------------|--------------------------------------------------|------------------------------------------------------------|
| Prompt injection       | Retrieved text says “reveal system prompt”       | Treat retrieved text as untrusted; strict tool permissions |
| Indirect injection     | Malicious instruction hidden in a web/PDF source | Sanitize/inspect sources; isolate instructions from data   |
| Data poisoning         | Attacker adds misleading documents               | Source trust, approvals, provenance, evaluation            |
| Unauthorized retrieval | Employee retrieves executive file                | ACL/tenant filtering before retrieval                      |
| Sensitive leakage      | Answer exposes PII                               | Authorization, redaction, output controls                  |
| Malicious documents    | Document exploits parser/model behavior          | Sandbox parsers, file validation, scanning                 |
| Output abuse           | Model generates unsafe action                    | Policy checks and least-privilege tools                    |

Security principle: retrieval authorization must be enforced by the application/data layer, not delegated to the LLM. An LLM should never be the final access-control mechanism.

## PART 31 — RAG vs FINE-TUNING

| Feature               | RAG                                      | Fine-Tuning                                          |
|-----------------------|------------------------------------------|------------------------------------------------------|
| Knowledge updates     | Easy: update/index documents             | Requires additional training/update cycle            |
| Private data          | Can retrieve controlled sources          | Can encode patterns, but data governance is critical |
| Cost                  | Inference/retrieval cost                 | Training + inference cost                            |
| Training              | No model training required               | Requires training data and process                   |
| Hallucination         | Can ground with evidence; not eliminated | Not inherently grounded                              |
| Style                 | Prompt-controlled; can also fine-tune    | Strong for persistent style/behavior                 |
| Domain knowledge      | Great for external facts                 | Can learn task/domain patterns                       |
| Real-time information | Strong fit with fresh index              | Poor fit for rapidly changing facts                  |

### When to use which

- Use RAG when the main problem is access to changing/private/factual information.
- Use fine-tuning when the main problem is behavior, format, style, task adaptation, or specialized response patterns.

- Use both when you need specialized behavior plus current/private knowledge.

## PART 32 — RAG vs OTHER SEARCH METHODS

| Method          | Best when                            | Weakness                       |
|-----------------|--------------------------------------|--------------------------------|
| Keyword search  | Exact words/phrases matter           | Poor paraphrase handling       |
| TF-IDF          | Simple lexical relevance             | Older/simple ranking           |
| BM25            | Strong lexical retrieval baseline    | Misses semantic paraphrases    |
| Dense retrieval | Semantic matching                    | Can miss exact identifiers     |
| Vector search   | Large-scale dense retrieval          | Requires embeddings/index      |
| Hybrid search   | Need lexical + semantic strength     | More components/tuning         |
| Knowledge graph | Explicit entities/relationships      | Graph construction/maintenance |
| GraphRAG        | Relationship/global corpus questions | Higher architecture complexity |

Practical rule: start with BM25 + dense retrieval as a strong baseline for mixed enterprise data, then add reranking and query transformation only when evaluation shows a problem.

## PART 33 — BEGINNER INTERVIEW QUESTIONS

**Q:** What is RAG?

**Answer:** A pattern that retrieves external evidence and gives it to an LLM as context before generation.

**Simple:** A pattern that retrieves external evidence and gives it to an LLM as context before generation. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** Why do we need RAG?

**Answer:** For current, private, large or domain-specific knowledge without retraining the model.

**Simple:** For current, private, large or domain-specific knowledge without retraining the model. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is an embedding?

**Answer:** A numeric vector representing semantic information in a learned space.

**Simple:** A numeric vector representing semantic information in a learned space. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is a vector database?

**Answer:** A system/index optimized for storing and searching vectors, often with metadata.

**Simple:** A system/index optimized for storing and searching vectors, often with metadata. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is chunking?

**Answer:** Splitting documents into retrievable units.

**Simple:** Splitting documents into retrievable units. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is chunk overlap?

**Answer:** Repeating a small boundary region between chunks to reduce lost context.

**Simple:** Repeating a small boundary region between chunks to reduce lost context. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is similarity search?

**Answer:** Finding vectors most similar to a query vector.

**Simple:** Finding vectors most similar to a query vector. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is top-K?

**Answer:** The K highest-ranked retrieved candidates.

**Simple:** The K highest-ranked retrieved candidates. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is metadata?

**Answer:** Structured information attached to a chunk, such as page, source, date or permissions.

**Simple:** Structured information attached to a chunk, such as page, source, date or permissions. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is hallucination?

**Answer:** A generated claim that is unsupported or incorrect.

**Simple:** A generated claim that is unsupported or incorrect. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is retrieval?

**Answer:** Selecting evidence relevant to a query.

**Simple:** Selecting evidence relevant to a query. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is dense retrieval?

**Answer:** Semantic retrieval using dense embeddings.

**Simple:** Semantic retrieval using dense embeddings. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is sparse retrieval?

**Answer:** Lexical retrieval using sparse term representations/ranking.

**Simple:** Lexical retrieval using sparse term representations/ranking. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is BM25?

**Answer:** A probabilistic lexical ranking function using term frequency, IDF and document length normalization.

**Simple:** A probabilistic lexical ranking function using term frequency, IDF and document length normalization. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is cosine similarity?

**Answer:** A measure of the angle-based similarity between vectors.

**Simple:** A measure of the angle-based similarity between vectors. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** Why not send the whole PDF to the LLM?

**Answer:** Context limits, cost, latency and noise make targeted retrieval preferable.

**Simple:** Context limits, cost, latency and noise make targeted retrieval preferable. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is reranking?

**Answer:** Reordering an initial candidate set using a stronger relevance model.

**Simple:** Reordering an initial candidate set using a stronger relevance model. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is a cross-encoder?

**Answer:** A model that jointly processes query and document to score relevance.

**Simple:** A model that jointly processes query and document to score relevance. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is hybrid retrieval?

**Answer:** Combining multiple retrieval signals, commonly BM25 and dense retrieval.

**Simple:** Combining multiple retrieval signals, commonly BM25 and dense retrieval. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is RRF?

**Answer:** A rank-fusion method that combines ranked lists using reciprocal rank.

**Simple:** A rank-fusion method that combines ranked lists using reciprocal rank. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is FAISS?

**Answer:** A library for efficient vector similarity search.

**Simple:** A library for efficient vector similarity search. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is HNSW?

**Answer:** A graph-based ANN index with hierarchical layers.

**Simple:** A graph-based ANN index with hierarchical layers. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is ANN?

**Answer:** Approximate nearest-neighbor search that trades some exactness for speed.

**Simple:** Approximate nearest-neighbor search that trades some exactness for speed. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is metadata filtering?

**Answer:** Restricting retrieval by structured attributes.

**Simple:** Restricting retrieval by structured attributes. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** Why are citations useful?

**Answer:** They let users verify the evidence supporting an answer.

**Simple:** They let users verify the evidence supporting an answer. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** Does RAG eliminate hallucination?

**Answer:** No; it can reduce unsupported generation but retrieval and generation can still fail.

**Simple:** No; it can reduce unsupported generation but retrieval and generation can still fail. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is indexing?

**Answer:** Preparing documents into chunks/vectors/metadata for fast retrieval.

**Simple:** Preparing documents into chunks/vectors/metadata for fast retrieval. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is query embedding?

**Answer:** The vector representation of the user's query in the same compatible space as document vectors.

**Simple:** The vector representation of the user's query in the same compatible space as document vectors. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is context?

**Answer:** The evidence placed into the LLM input for answering the query.

**Simple:** The evidence placed into the LLM input for answering the query. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

**Q:** What is a retriever?

**Answer:** The component/interface that returns relevant documents for a query.

**Simple:** The component/interface that returns relevant documents for a query. Example: a company bot retrieves the HR handbook before answering a leave question.

**Common mistake:** Treating the term as magic instead of explaining its place in the pipeline.

## PART 34 — INTERMEDIATE INTERVIEW QUESTIONS

**Q:** Dense vs sparse retrieval?

**Answer:** Dense captures semantic similarity; sparse such as BM25 captures lexical overlap. Hybrid often gives stronger coverage.

**Q:** Why BM25 over simple keyword search?

**Answer:** It models term importance, TF saturation and document length normalization.

**Q:** Why cosine similarity?

**Answer:** It compares vector direction and is commonly aligned with semantic embedding spaces.

**Q:** What is ANN trade-off?

**Answer:** Lower query work and latency in exchange for potentially lower recall.

**Q:** FAISS vs vector DB?

**Answer:** FAISS is a similarity-search library; vector DBs add persistence, metadata, APIs, operational and scaling capabilities.

**Q:** How does HNSW search work?

**Answer:** Navigate a hierarchical proximity graph from upper layers to the detailed base layer.

**Q:** What is M in HNSW?

**Answer:** A graph connectivity parameter affecting memory, build time and search quality.

**Q:** What is efSearch?

**Answer:** Query-time exploration breadth controlling recall/latency.

**Q:** How do you choose chunk size?

**Answer:** Start with document/task-aware ranges, then tune using retrieval metrics and qualitative failures.

**Q:** Why can larger chunks hurt?

**Answer:** They mix unrelated topics, making embeddings less focused and context more expensive.

**Q:** Why can smaller chunks hurt?

**Answer:** They may omit definitions or prerequisites needed to interpret a fact.

**Q:** Why metadata filtering?

**Answer:** Accuracy, routing, freshness and especially authorization.

**Q:** What is pre-filtering?

**Answer:** Restrict candidates before similarity search.

**Q:** What is reranking?

**Answer:** Use a stronger relevance model to reorder a smaller candidate set.

**Q:** Bi-encoder vs cross-encoder?

**Answer:** Bi-encoder is scalable and precomputable; cross-encoder is slower but more precise.

**Q:** What is RRF useful for?

**Answer:** Combining rankings from heterogeneous retrievers without score calibration.

**Q:** What is hybrid search?

**Answer:** Combining lexical and semantic retrieval.

**Q:** What is query expansion?

**Answer:** Adding/reformulating terms to improve recall.

**Q:** What is multi-query?

**Answer:** Generate multiple query formulations and merge retrieval results.

**Q:** What is HyDE?

**Answer:** Retrieve using an embedding of a hypothetical generated document.

**Q:** What is parent-child retrieval?

**Answer:** Retrieve small child chunks then expand to a larger parent context.

**Q:** What is contextual retrieval?

**Answer:** Add disambiguating context to chunks before indexing/retrieval.

**Q:** Why deduplicate chunks?

**Answer:** Overlap and multiple retrievers can return the same evidence repeatedly.

**Q:** What is context window pressure?

**Answer:** Large context consumes tokens, cost and latency and can reduce effective attention.

**Q:** Why evaluate retrieval separately?

**Answer:** Generation metrics alone cannot tell whether the right evidence was retrieved.

**Q:** Precision vs recall in retrieval?

**Answer:** Precision measures correctness among retrieved items; recall measures coverage of relevant items.

**Q:** What is MRR?

**Answer:** Average reciprocal rank of the first relevant result.

**Q:** What is NDCG?

**Answer:** A normalized ranked-list metric that rewards highly relevant results appearing early.

**Q:** What is faithfulness?

**Answer:** Whether generated claims are supported by retrieved context.

**Q:** What is answer relevance?

**Answer:** Whether the answer addresses the query.

**Q:** What is context relevance?

**Answer:** Whether retrieved evidence is relevant to the query.

**Q:** Why can a high retrieval score still produce a bad answer?

**Answer:** The reranker may select noisy evidence, the prompt may be poor, or the LLM may misread/ignore context.

**Q:** Why use query rewriting?

**Answer:** Natural user questions may be poor search queries; rewriting can improve retrieval.

**Q:** When is BM25 especially useful?

**Answer:** Exact product IDs, error codes, names, legal clauses and rare terminology.

**Q:** Why not use only embeddings?

**Answer:** Embeddings can miss exact lexical constraints and identifiers.

**Q:** Why not use only BM25?

**Answer:** Lexical matching misses paraphrases and semantic relationships.

**Q:** What is a retriever interface?

**Answer:** An abstraction that maps a query to relevant documents, hiding storage/index details.

**Q:** Why is top-K tuning important?

**Answer:** It controls recall, noise, token usage and latency.

**Q:** What is context compression?

**Answer:** Reduce retrieved evidence to the parts relevant to the query while preserving support.

**Q:** Why use a reranker after hybrid retrieval?

**Answer:** Fusion improves candidate coverage; reranking improves final ordering.

## PART 35 — ADVANCED INTERVIEW QUESTIONS

### 1. How would you improve poor retrieval accuracy?

Build a labeled evaluation set; inspect misses; test chunking, embeddings, query rewriting, hybrid retrieval, filters, candidate K and reranking; measure recall/precision rather than guessing.

### 2. How do you select chunk size?

Use document structure and answer granularity as hypotheses, then benchmark multiple sizes against retrieval and end-to-end metrics.

### 3. How do you choose top-K?

Tune K against recall, reranking quality, context length, latency and cost. Often retrieve broadly then rerank narrowly.

#### **4. Why can larger chunks hurt?**

Embedding becomes a mixture of topics and the relevant sentence can be diluted.

#### **5. Why can smaller chunks lose context?**

Definitions, headings, table headers and references may be separated.

#### **6. How debug hallucinations?**

Trace retrieved chunks, verify evidence coverage, check prompt, inspect unsupported claims, then determine retrieval vs generation failure.

#### **7. How evaluate retrieval independently?**

Create queries with known relevant documents/chunks and compute recall@K, precision@K, hit rate, MRR and NDCG.

#### **8. Why rerank?**

Fast first-stage retrieval optimizes recall; reranking spends compute on a small candidate set to improve precision.

#### **9. Why hybrid retrieval?**

Dense and lexical methods have complementary failure modes.

#### **10. When GraphRAG?**

When questions depend on entities/relationships or global corpus structure rather than isolated passages.

#### **11. When Agentic RAG?**

When answering requires planning, multiple tools/retrieval steps or iterative evidence gathering.

#### **12. How does HNSW work?**

A hierarchical proximity graph supports greedy navigation and broader candidate exploration near the base layer.

#### **13. What is ANN trade-off?**

Speed/memory vs exactness/recall.

#### **14. How scale vector search?**

ANN indexes, partitioning/sharding, replicas, metadata-aware routing, efficient embeddings and capacity planning.

#### **15. How reduce RAG latency?**

Profile stages; parallelize retrieval sources; cache embeddings/results; use ANN; reduce candidate/reranker sizes; stream generation.

#### **16. How reduce cost?**

Reduce redundant retrieval/LLM calls, compress context, cache, batch indexing, route simple queries to cheaper models.

#### **17. How handle changing documents?**

Version documents, incrementally index updates, remove stale vectors, and track source timestamps.

**18. How handle deletion?**

Delete all chunks/vectors/derived caches and ensure replicas and asynchronous pipelines converge.

**19. How implement access control?**

Carry tenant/user/role metadata and enforce filters before unauthorized content can enter the model context.

**20. How prevent prompt injection?**

Treat retrieved text as untrusted data, separate instructions from data, restrict tools, validate actions and avoid giving the model broad privileges.

**21. How prevent data poisoning?**

Source trust, ingestion validation, provenance, approvals, anomaly detection and evaluation against trusted references.

**22. How handle millions of vectors?**

ANN, memory/index tuning, partitioning, sharding and replicas; measure recall/latency.

**23. How compare embedding models?**

Benchmark on representative queries using recall@K, MRR/NDCG, latency, vector size and cost.

**24. How compare rerankers?**

Measure ranking metrics and end-to-end answer quality against latency/cost.

**25. What is retrieval recall ceiling?**

If the correct evidence is not retrieved, a grounded generator cannot reliably use it; generation cannot recover missing evidence unless it has the fact independently.

**26. Why is retrieval quality not enough?**

Correct evidence can be retrieved but ignored or contradicted by the generator.

**27. How design a RAG evaluation set?**

Representative queries, known source evidence, relevance labels and expected answer/reference where needed; include adversarial and no-answer cases.

**28. How handle no-answer questions?**

Detect insufficient evidence and instruct the system to abstain rather than invent.

**29. What is query routing?**

Choose a retrieval/tool path based on question type, such as SQL for structured metrics and vector search for documents.

**30. How combine SQL and RAG?**

Route structured questions to SQL and unstructured questions to retrieval, or combine outputs when the task genuinely requires both.

**31. How avoid duplicate context?**

Canonical chunk IDs, overlap-aware deduplication, semantic similarity checks and source-level deduplication.

### **32. What is lost-in-the-middle?**

Models may use information less effectively when critical evidence is buried in a very long context; selecting and ordering evidence can help.

### **33. How monitor production RAG?**

Trace retrieval scores, candidate IDs, final context, latency, token usage, model versions and evaluation signals while protecting sensitive data.

### **34. Why version indexes?**

Embedding/index changes can alter retrieval behavior; versioning enables rollback and controlled evaluation.

### **35. How handle multilingual RAG?**

Use multilingual embeddings/retrieval and test each target language; hybrid lexical retrieval may require language-aware analyzers.

### **36. How handle PDFs with tables?**

Use structure-aware extraction and preserve table headers/rows; plain text extraction can destroy relationships.

### **37. How handle duplicate documents?**

Detect duplicates at ingestion using hashes/content similarity and retain canonical source IDs.

### **38. How design a safe web fallback?**

Allow only authorized domains/tools, treat web text as untrusted, cite sources and enforce output/tool policies.

### **39. What is retrieval calibration?**

Understanding score distributions and thresholds so the system can decide whether evidence is sufficiently relevant.

### **40. How do you decide whether to add a technique?**

Only add it when an evaluation or production failure demonstrates a problem it is likely to solve; complexity has an operational cost.

## **PART 36 — SCENARIO-BASED INTERVIEW QUESTIONS**

### **Scenario 1: Company chatbot gives irrelevant answers**

Inspect chunking → embeddings → candidate recall → metadata filters → top-K → reranking → context construction → prompt. Measure each stage.

### **Scenario 2: Chatbot hallucinates despite correct retrieval**

Verify the correct evidence is in final context; inspect prompt and unsupported claims; add grounding/abstention behavior and evaluation.

### **Scenario 3: Latency is 5 seconds**

Trace ingestion/query stages; isolate embedding, retrieval, reranker, LLM/network. Optimize the slowest stage rather than guessing.

#### **Scenario 4: Cost doubled after launch**

Check query volume, context tokens, model calls, reranker calls and cache hit rate. Reduce redundant work and route appropriately.

#### **Scenario 5: Corpus grows to 10 million chunks**

Move from brute force to ANN; consider partitioning/sharding, replicas, metadata routing and memory/index tuning.

#### **Scenario 6: Documents change every hour**

Use incremental indexing, versioning and stale-vector deletion rather than rebuilding everything.

#### **Scenario 7: Multi-tenant SaaS RAG**

Every chunk carries tenant identity; enforce tenant filters at retrieval and test cross-tenant leakage.

#### **Scenario 8: Sensitive executive documents**

Use ACL-aware pre-filtering and separate security boundaries; never rely on the LLM to decide authorization.

#### **Scenario 9: Exact error codes are missed**

Add BM25/hybrid retrieval because lexical exact matches matter.

#### **Scenario 10: Paraphrased questions are missed**

Test dense retrieval, query rewriting and embedding model quality.

#### **Scenario 11: Top-20 contains useful docs but final answer is wrong**

Inspect reranking and final context selection; the useful evidence may be pushed out.

#### **Scenario 12: PDF tables produce nonsense**

Use table-aware extraction and chunking; preserve headers and row relationships.

#### **Scenario 13: Long legal contracts**

Use section-aware chunking, parent-child retrieval, metadata for clause types, and citations.

#### **Scenario 14: Need relationship-heavy research**

Consider GraphRAG/knowledge graph if the key task is entity traversal rather than passage similarity.

#### **Scenario 15: One question requires five searches**

Agentic RAG can orchestrate iterative retrieval, but constrain tools and evaluate each step.

#### **Scenario 16: No relevant evidence exists**

Implement abstention/no-answer behavior and evaluate false-confidence cases.

#### **Scenario 17: Embedding model upgraded**

Create a new versioned index, benchmark against old, migrate gradually and keep rollback capability.

#### **Scenario 18: User asks for a document they are not allowed to see**

Authorization filter must block retrieval before context construction.

#### **Scenario 19: Malicious PDF contains instructions**

Treat content as data; do not execute instructions from documents; sandbox parsers and restrict tools.

#### **Scenario 20: Retriever has high recall but low precision**

Rerank, improve chunking, add metadata filters, or tune candidate/final K.

### Scenario 21: High precision but low recall

Increase candidate K, improve query transformation, add dense/hybrid retrieval or adjust chunking.

### Scenario 22: Answers are too verbose

Use a context/output budget and prompt for concise answers while retaining citations.

### Scenario 23: Answers cite wrong pages

Preserve stable page/chunk metadata and test citation mapping end-to-end.

### Scenario 24: Duplicate sources dominate context

Deduplicate by document/chunk identity and semantic similarity.

### Scenario 25: Freshness matters more than semantic similarity

Add timestamp/recency metadata and retrieval policy; do not let old high-similarity chunks win automatically.

### Scenario 26: Search must support exact product SKUs

Use lexical/BM25 retrieval or hybrid search.

### Scenario 27: RAG works on English but not Hindi

Benchmark multilingual embeddings and language-specific lexical analyzers; evaluate Hindi queries separately.

### Scenario 28: Need offline/on-prem deployment

Choose local embeddings/LLM and FAISS/Qdrant/pgvector based on scale and operational constraints; ensure model licenses permit use.

### Scenario 29: Evaluation scores are high but users complain

Check whether the evaluation set represents real traffic; add real user queries and failure categories.

## PART 37 — CODING INTERVIEW QUESTIONS

### 1. Cosine similarity

```
import numpy as np

def cosine_similarity(a, b):
    a = np.asarray(a, dtype=float)
    b = np.asarray(b, dtype=float)
    denom = np.linalg.norm(a) * np.linalg.norm(b)
    if denom == 0:
        return 0.0
    return np.dot(a, b) / denom
```

Complexity:  $O(d)$  for dimension  $d$ . Interview follow-up: what changes for many documents? Use an index/ANN rather than looping over all vectors.

### 2. Top-K retrieval

```
def top_k(query, vectors, k):
    scores = [cosine_similarity(query, v) for v in vectors]
    ranked = sorted(enumerate(scores), key=lambda x: x[1], reverse=True)
    return ranked[:k]
```

Naive complexity:  $O(n \cdot d + n \log n)$  due to scoring and full sorting. A heap can reduce selection cost to  $O(n \log k)$  after scoring.

### 3. Vector normalization

```
def normalize(v):
    norm = np.linalg.norm(v)
    return v if norm == 0 else v / norm
```

Unit normalization makes dot product equal to cosine similarity.

### 4. Simple chunker

```
def chunk_text(text, size=500, overlap=50):
    chunks = []
    start = 0
    while start < len(text):
        end = min(len(text), start + size)
        chunks.append(text[start:end])
        if end == len(text):
            break
        start = end - overlap
    return chunks
```

This is a character-based teaching example, not a production tokenizer-aware splitter.

### 5. Metadata filtering

```
def filter_docs(docs, department, year):
    return [
        d for d in docs
        if d["metadata"]["department"] == department
        and d["metadata"]["year"] == year
    ]
```

### 6. Simple reranking

```
def rerank(candidates, score_fn, top_n=5):
    scored = [(doc, score_fn(doc)) for doc in candidates]
    scored.sort(key=lambda x: x[1], reverse=True)
    return [doc for doc, _ in scored[:top_n]]
```

### 7. Basic NumPy vector search

```
def search(query, matrix, k=5):
    q = query / np.linalg.norm(query)
    X = matrix / np.linalg.norm(matrix, axis=1, keepdims=True)
    scores = X @ q
    idx = np.argsort(scores)[::-1][:k]
    return idx, scores[idx]
```

With normalized rows, matrix-vector multiplication computes cosine similarities. Complexity is  $O(n \cdot d)$  for brute-force scoring.

### 8. Basic RAG pipeline

```
def rag(query):
    q_vec = embed(query)
    candidates = retrieve(q_vec, index, k=20)
    candidates = apply_acl_and_metadata(candidates)
    candidates = rerank(query, candidates)[:5]
    context = format_context(candidates)
    prompt = build_grounded_prompt(query, context)
    answer = llm(prompt)
    return answer, citations(candidates)
```

## PART 38 — MATHEMATICS REQUIRED FOR RAG

### Vectors

A vector is an ordered list of numbers. In RAG, an embedding might be represented as [0.2, -0.1, 0.8]. Each coordinate is not usually interpretable alone; meaning is distributed across dimensions.

### Dot product

$$a \cdot b = a_1b_1 + a_2b_2 + \dots + a_db_d$$

Example:  $[1,2] \cdot [3,4] = 1 \times 3 + 2 \times 4 = 11$ .

### Magnitude

$$||a|| = \sqrt{a_1^2 + a_2^2 + \dots + a_d^2}$$

For [3,4], magnitude = 5.

### Cosine similarity

$$\cos(a,b) = (a \cdot b) / (||a|| ||b||)$$

Example [1,0] and [0,1]  $\rightarrow$  0. [1,0] and [1,0]  $\rightarrow$  1. [1,0] and [-1,0]  $\rightarrow$  -1.

### Euclidean distance

$$d(a,b) = \sqrt{\sum (a_i - b_i)^2}$$

For [1,2] and [4,6], distance =  $\sqrt{9+16}=5$ .

### TF and IDF

TF measures how often a term appears in a document. IDF measures how uncommon/discriminative the term is across documents. A common simplified IDF is  $\log(N/df)$ , where N is number of documents and df is number containing the term.

### BM25

BM25 adds term-frequency saturation and document-length normalization to lexical retrieval. In interviews, explain the purpose of each component before writing the formula.

### Precision and recall

$$\text{Precision} = TP / (TP + FP)$$

$$\text{Recall} = TP / (TP + FN)$$

In retrieval, TP means a retrieved relevant item, FP a retrieved irrelevant item, and FN a relevant item that was not retrieved.

### MRR, DCG, NDCG

$$RR = 1 / \text{rank}(\text{first relevant})$$

$$MRR = \text{average}(RR)$$

$$DCG@k = \sum (2^{\text{rel}_i} - 1) / \log_2(i+1)$$

$$NDCG@k = DCG@k / IDC@k$$

Practical use: these metrics quantify whether the evidence is not just present, but placed high enough in the ranked list to be useful.

## PART 39 — CHEAT SHEETS

### RAG 1-page cheat sheet

#### INGESTION:

documents → parse → clean → chunk → metadata → embed → index

#### QUERY:

query → preprocess → embed/rewrite → retrieve → filter → rerank  
→ context → prompt → LLM → answer + citations

#### KEY QUESTIONS:

Did we retrieve the right evidence?

Did we put the right evidence in context?

Did the model answer only from/support by that evidence?

Can the user verify the source?

Can unauthorized data ever enter the context?

### Retrieval cheat sheet

| Term          | Remember                                           |
|---------------|----------------------------------------------------|
| BM25          | Lexical relevance: TF + IDF + length normalization |
| Dense         | Semantic vector retrieval                          |
| Vector search | Nearest vectors to query vector                    |
| ANN           | Fast approximate nearest-neighbor search           |
| FAISS         | Similarity-search library                          |
| HNSW          | Graph-based ANN index                              |
| Reranker      | Second-stage relevance model                       |
| Cross-encoder | Joint query-document scoring model                 |
| RRF           | Fuse ranked lists                                  |
| Hybrid        | Dense + lexical retrieval                          |

### Architecture cheat sheet

Simple: Query → Retrieve → Generate

Advanced: Query → Transform → Hybrid Retrieve → Filter  
→ Rerank → Context → Generate

Agentic: Query → Plan ↔ Tools/Retrievers → Synthesize

GraphRAG: Query → Entity/Graph retrieval → Evidence → Generate

### 50 rapid-fire interview answers

**Q:** RAG?

**Answer:** Retrieve evidence, then generate.

**Q:** Embedding?

**Answer:** Semantic vector representation.

**Q:** Vector DB?

**Answer:** Stores/searches vectors plus metadata.

**Q:** Chunking?

**Answer:** Makes documents retrievable.

**Q:** Overlap?

**Answer:** Reduces boundary loss.

**Q:** Top-K?

**Answer:** Number of candidates returned.

**Q:** BM25?

**Answer:** Strong lexical ranking baseline.

**Q:** Dense retrieval?

**Answer:** Embedding-based semantic search.

**Q:** Hybrid?

**Answer:** Combine lexical + dense.

**Q:** Cosine?

**Answer:** Angle-based vector similarity.

**Q:** ANN?

**Answer:** Faster approximate neighbor search.

**Q:** FAISS?

**Answer:** Vector similarity library.

**Q:** HNSW?

**Answer:** Graph-based ANN.

**Q:** Reranking?

**Answer:** Second-stage candidate ordering.

**Q:** Cross-encoder?

**Answer:** Joint query-doc relevance model.

**Q:** RRF?

**Answer:** Rank fusion.

**Q:** Metadata?

**Answer:** Structured chunk attributes.

**Q:** ACL?

**Answer:** Authorization controlling access.

**Q:** Recall?

**Answer:** Relevant evidence retrieved.

**Q:** Precision?

**Answer:** Retrieved evidence that is relevant.

**Q:** MRR?

**Answer:** First relevant rank quality.

**Q:** NDCG?

**Answer:** Graded ranked-list quality.

**Q:** Faithfulness?

**Answer:** Answer supported by context.

**Q:** Answer relevance?

**Answer:** Answer addresses query.

**Q:** Context relevance?

**Answer:** Retrieved evidence is useful.

**Q:** HyDE?

**Answer:** Retrieve from hypothetical-document embedding.

**Q:** Multi-query?

**Answer:** Multiple query formulations.

**Q:** Parent-child?

**Answer:** Retrieve small, expand larger context.

**Q:** Contextual retrieval?

**Answer:** Add disambiguating chunk context.

**Q:** CRAG?

**Answer:** Evaluate/correct retrieval.

**Q:** Self-RAG?

**Answer:** Adaptive retrieval/reflection.

**Q:** Agentic RAG?

**Answer:** Planned multi-step tool retrieval.

**Q:** GraphRAG?

**Answer:** Graph/relationship-aware retrieval.

**Q:** Indexing?

**Answer:** Prepare searchable representations.

**Q:** Query embedding?

**Answer:** Vector for user query.

**Q:** Chunk embedding?

**Answer:** Vector for source chunk.

**Q:** Grounding?

**Answer:** Tie answer to evidence.

**Q:** Abstention?

**Answer:** Say insufficient evidence when needed.

**Q:** No-answer case?

**Answer:** Important evaluation category.

**Q:** Prompt injection?

**Answer:** Untrusted text tries to control model.

**Q:** Data poisoning?

**Answer:** Bad source data contaminates retrieval.

**Q:** Embedding versioning?

**Answer:** Keep vector spaces/indexes consistent.

**Q:** Incremental indexing?

**Answer:** Update only changed data.

**Q:** Deletion?

**Answer:** Remove source and all derived representations.

**Q:** Latency?

**Answer:** Measure each pipeline stage.

**Q:** Cost?

**Answer:** Control model calls and tokens.

**Q:** Lost in middle?

**Answer:** Long context can reduce use of buried evidence.

**Q:** Query routing?

**Answer:** Choose the best backend/tool.

**Q:** Source attribution?

**Answer:** Return document/page evidence.

**Q:** Production priority?

**Answer:** Measure before adding complexity.

## PART 40 — A–Z TERMINOLOGY GLOSSARY

### ACL

Access Control List governing who can access a resource.

### ANN

Approximate Nearest Neighbor — fast search for near vectors without exact comparison to every vector.

### Agentic RAG

RAG where an agent plans and performs multiple retrieval/tool steps.

### Answer Relevancy

How well the generated answer addresses the query.

### BM25

A lexical ranking function using term frequency, inverse document frequency and length normalization.

### CRAG

Corrective RAG; retrieval quality is assessed and corrected.

### Chunk

A retrievable unit of source content.

### Chunk Overlap

Repeated boundary tokens/characters across adjacent chunks.

### Context

Evidence supplied to the LLM for a query.

### Context Precision

How highly relevant retrieved context is ranked.

### Context Recall

How much required evidence was retrieved.

### Contextual Retrieval

Attach disambiguating context to chunks before retrieval.

### Cosine Similarity

Dot product divided by vector magnitudes.

### Cross-encoder

A model that jointly processes query and document to score relevance.

### Data Poisoning

Malicious or erroneous data contaminating the knowledge source.

### Dot Product

Sum of coordinate-wise products.

### Embedding

A learned numeric vector representation.

## **Embedding Dimension**

Number of coordinates in an embedding vector.

## **Euclidean Distance**

Straight-line distance between vectors.

## **FAISS**

A library for efficient similarity search over vectors.

## **Faithfulness**

Degree to which generated claims are supported by context.

## **Freshness**

How current the indexed information is.

## **GraphRAG**

RAG using graph structure/entities/relationships.

## **Grounding**

Constraining/supporting generation using external evidence.

## **HNSW**

Hierarchical Navigable Small World graph index for ANN search.

## **HyDE**

Hypothetical Document Embeddings; use generated hypothetical text for retrieval.

## **Hybrid Search**

Combining multiple retrieval methods, commonly dense and BM25.

## **IDF**

Inverse document frequency; rewards terms that occur in fewer documents.

## **IVF**

Inverted File index that partitions vectors into coarse clusters.

## **Index**

Data structure representing searchable content efficiently.

## **Indexing Pipeline**

Process that converts source documents into searchable records.

## **Indirect Prompt Injection**

Injection embedded in retrieved external content.

## **LLM**

Large Language Model used for generation/reasoning.

## **MRR**

Mean Reciprocal Rank.

## **Metadata**

Structured attributes attached to source chunks.

## **Metadata Filter**

Structured predicate limiting retrieval candidates.

## **Multi-Query**

Generate multiple query variants and merge results.

## **Multi-tenancy**

Serving multiple customers while isolating their data.

## **NDCG**

Normalized Discounted Cumulative Gain.

## **No-answer / Abstention**

Explicitly declining to answer when evidence is insufficient.

## **Normalization**

Scaling a vector, often to unit length.

## **PII**

Personally identifiable information requiring careful protection.

## **PQ**

Product Quantization for vector compression.

## **Parent-Child Retrieval**

Retrieve fine-grained children and expand to parent context.

## **Precision**

Relevant retrieved items divided by all retrieved items.

## **Prompt Injection**

Untrusted text attempts to alter model instructions.

## **Query Expansion**

Adding related terms/reformulations to improve retrieval.

## **Query Pipeline**

Online path from user query to final answer.

## **RAG**

Retrieval-Augmented Generation.

## **RRF**

Reciprocal Rank Fusion for combining ranked lists.

## **Recall**

Relevant retrieved items retrieved divided by all relevant items.

## Reranker

Second-stage model that reorders retrieved candidates.

## Reranking

Reorder a candidate set with a more accurate scoring model.

## Retriever

Component that returns relevant source items.

## TF

Term frequency; count/weight of a term in a document.

## Top-K

The K highest-ranked retrieved candidates.

## Vector DB

Database/system optimized for vector storage and search.

## Vector Index

Data structure accelerating vector search.

## Vector Search

Nearest-neighbor search in vector space.

# PART 41 — FINAL LEARNING ROADMAP

```
Level 1 → RAG Basics
↓
Level 2 → Embeddings + Vector Search
↓
Level 3 → Retrieval
↓
Level 4 → Advanced Retrieval
↓
Level 5 → Advanced RAG
↓
Level 6 → Evaluation
↓
Level 7 → Production RAG
↓
Level 8 → Interview Preparation
```

| Level                                | Learn                                                                            | Practice                                                          | Interview goal                   |
|--------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------|
| Level 1 — RAG Basics                 | Learn RAG, LLM limitations, indexing/query split, chunks, metadata.              | Draw the RAG pipeline from memory; build a toy keyword retriever. | Explain RAG vs LLM/fine-tuning.  |
| Level 2 — Embeddings + Vector Search | Vectors, embeddings, cosine, dot product, vector DB, FAISS, ANN.                 | Implement cosine and NumPy top-K; index a small document set.     | Explain embedding space and ANN. |
| Level 3 — Retrieval                  | BM25, dense retrieval, hybrid search, top-K, filtering.                          | Compare BM25 and dense retrieval on 30–50 queries.                | Precision/recall, BM25, hybrid.  |
| Level 4 — Advanced Retrieval         | Reranking, cross-encoder, RRF, query expansion, multi-query, HyDE, parent-child. | Add a reranker and measure                                        | Why reranking/hybrid?            |

## RAG Mastery Guide — From Zero to Interview Ready

|                          |                                                                              |                                                                               |                                         |
|--------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------|
|                          |                                                                              | recall@K/MRR.                                                                 |                                         |
| Level 5 — Advanced RAG   | CRAG, Self-RAG, Agentic RAG, GraphRAG, routing.                              | Build one multi-step/graph-oriented prototype only if your use case needs it. | Trade-offs and architecture selection.  |
| Level 6 — Evaluation     | Retrieval metrics, faithfulness, answer relevance, RAGAS-style evaluation.   | Create an evaluation set and failure taxonomy.                                | Debug retrieval vs generation.          |
| Level 7 — Production RAG | Security, ACLs, multi-tenancy, updates, deletion, latency, cost, monitoring. | Dockerize PDF QA with versioned indexes and access controls.                  | Design a production architecture.       |
| Level 8 — Interview      | Beginner/intermediate/advanced/scenario/coding practice.                     | Explain your project on a whiteboard and implement retrieval from scratch.    | Be able to justify every design choice. |

### Final checklist: can you do these?

- Explain RAG from scratch in 60 seconds and 5 minutes.
- Draw offline indexing and online query pipelines.
- Explain chunk size/overlap and tune them experimentally.
- Explain embeddings, dimensions, normalization and similarity.
- Implement cosine similarity and top-K retrieval.
- Explain BM25 and why it complements dense retrieval.
- Explain ANN, FAISS, IVF, PQ and HNSW at interview depth.
- Design hybrid retrieval with reranking.
- Explain RRF, cross-encoders, HyDE and parent-child retrieval.
- Compare Naive, Advanced, Modular, CRAG, Self-RAG, Agentic RAG and GraphRAG.
- Evaluate retrieval with precision, recall, hit rate, MRR and NDCG.
- Evaluate generation with faithfulness and answer/context relevance.
- Build a PDF QA RAG system with citations.
- Explain LangChain without hiding behind the framework.
- Design production security: ACLs, prompt injection defenses, PII and multi-tenancy.
- Handle updates, deletion, embedding versioning and stale data.
- Debug poor retrieval systematically.
- Answer at least 30 beginner, 40 intermediate, 40 advanced and 30 scenario questions.

### 30-second interview summary

RAG is a retrieval-and-generation architecture. During ingestion, documents are parsed, cleaned, chunked, enriched with metadata, embedded and indexed. At query time, the system embeds or transforms the query, retrieves candidates using dense, sparse or hybrid search, applies filters and often reranks them, builds a compact evidence context, and asks an LLM to generate a grounded response with citations. Production RAG is mainly an information-retrieval and systems-engineering problem: retrieval quality, security, freshness, latency, cost, evaluation and observability matter as much as the LLM.

### Final principle

Do not add complexity because a technique sounds advanced. Start with a measurable baseline, identify the failure, choose the smallest technique likely to fix it, and re-evaluate. Strong RAG engineering is evidence-driven.

## About this guide

This guide was generated from the uploaded RAG mastery specification. The specification explicitly asks for a standalone resource progressing from fundamentals to advanced/production/interview topics, with diagrams, tables, formulas, numerical examples, Python code, comparisons, common mistakes and interview preparation.

Coverage follows the requested organization and terminology. Some library APIs, model names and evaluation-framework APIs change over time; implementation examples should therefore be checked against the versions installed in your project.

Source specification citation: the uploaded request defines the beginner-to-interview goal and teaching style.

It also requires the final resource to be structured, technically accurate, interview-oriented, and include diagrams, tables, formulas, numerical examples, Python code, comparisons, cheat sheets and a glossary.